Distributed Database Concepts

Introduction

- Distributed computing system
  - Consists of several processing sites or nodes interconnected by a computer network
  - Nodes cooperate in performing certain tasks
  - Partitions large task into smaller tasks for efficient solving
- Big data technologies
  - Combine distributed and database technologies
  - Deal with mining vast amounts of data

Distributed Database Concepts

- What constitutes a distributed database?
  - Connection of database nodes over computer network
  - Logical interrelation of the connected databases
  - Possible absence of homogeneity among connected nodes
- Distributed database management system (DDBMS)
  - Software system that manages a distributed database

Distributed Database Management System(DDBMS)

- A system that is used for managing the storage and retrieval of data across multiple interconnected databases “Distributed db”.
- In this case, the interconnected databases are situated in different geographical areas.
- In DDBMS, one can access and store data transparently from different geographical locations and ensure high availability, scalability, and fault tolerance mechanisms.
- DDBMSs are designed to handle huge amounts of data spread across different sites. It can give seamless performance in data sharing and collaboration by organizations.